WEEK 9 – HOUSING DATA CLEANING AND TRANSFORMATION

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1. INTRODUCTION

Data cleaning is an important process in data analysis. It helps to identify and correct errors, remove duplicate records, handle missing or unwanted values, and improve data quality.

In this project, the King County House Sales Dataset is cleaned and transformed using Microsoft Power BI and Power Query Editor. Various data cleaning techniques and calculated columns are applied to prepare the dataset for further analysis

2. OBJECTIVES

- To clean and prepare the housing dataset for analysis.
- To identify and remove duplicate records.
- To filter unwanted or invalid values.
- To extract Year, Month and Day from the date column.
- To create house size and price categories.
- To convert waterfront values into meaningful labels.
- To create new calculated columns for better analysis.

3. DATASET DESCRIPTION

Dataset Name: King County House Sales Dataset

The dataset contains information about house sales in King County.

Important columns include:

- ID – Unique identification number of each property.
- Date – Date of house sale.
- Price – Selling price of the house.
- Bedrooms – Number of bedrooms.
- Bathrooms – Number of bathrooms.
- Sqft Living – Living area of the house.
- Sqft Lot – Total land area.
- Waterfront – Indicates whether the property has waterfront access.
- View – Rating of the property view.
- Grade – Overall construction and design grade.
- Condition – Overall condition of the house.
- Yr Built – Year the house was constructed.
- Yr Renovated – Year of renovation.
- Lat and Long – Geographic location of the property.

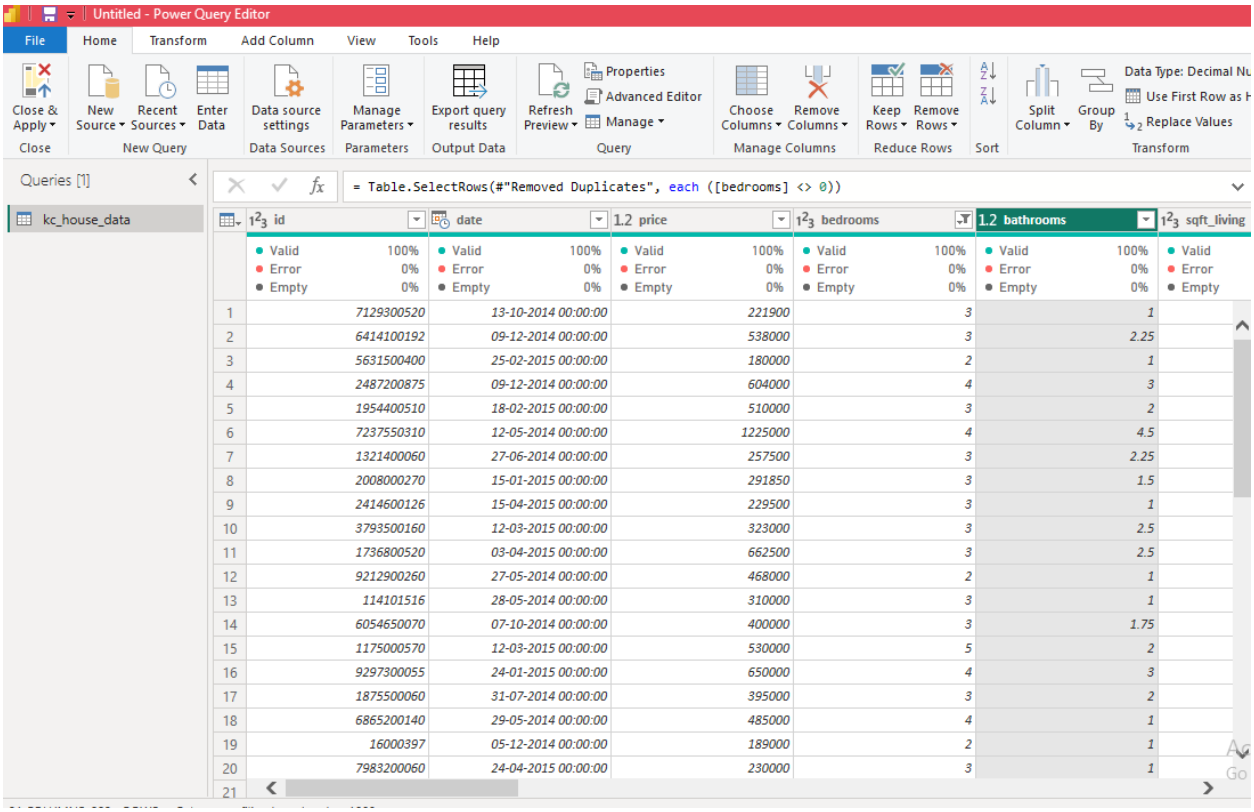
4.TOOLS AND TECHNOLOGIES USED

- Microsoft Power BI

- Power Query Editor
- Data Transformation
- Custom Columns
- Conditional Columns

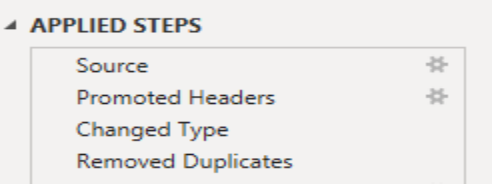
5.1 DATA CLEANING PROCESS

The King County House Sales Dataset was imported into Microsoft Power BI. The dataset was opened in Power Query Editor to perform the required cleaning and transformation operations.



5.2 Removing Duplicate Records

The Remove Duplicates option was used to identify and remove duplicate records from the dataset. This process helps avoid repeated entries and improves data accuracy.

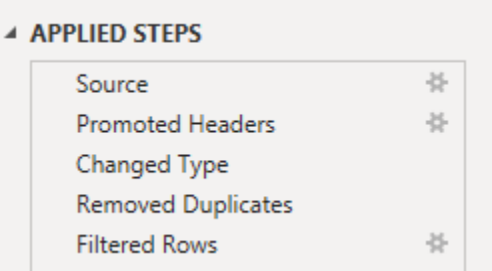


5.3 Filtering Unwanted Rows

The bedroom column was checked, and records containing zero bedrooms were filtered out.

The bathroom column was also checked for zero values. No zero values were visible in the available dropdown list.

This process helps remove unsuitable records from the analysis.

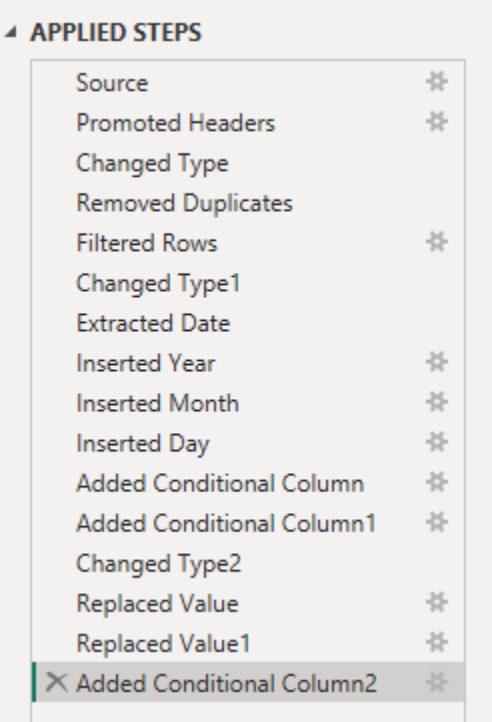


5.4 Changing Data Types

The data types of different columns were checked and modified according to their values.

Examples include:

- Date column – Date
- Price column – Decimal Number
- Bedrooms – Whole Number
- Bathrooms – Decimal Number
- Waterfront – Text



6. DATE EXTRACTION

6.1 Extracting Year

A new column named `Year` was created by extracting the year from the original date column.

This helps analyze house sales based on the year of transaction.

6.2 Extracting Month

A new column named `Month` was created to identify the month in which each house was sold.

6.3 Extracting Day

A new column named `Day` was created to identify the day of the transaction.

These columns make it easier to perform time-based analysis.

7. CREATING CONDITIONAL COLUMNS

7.1 House Size Category

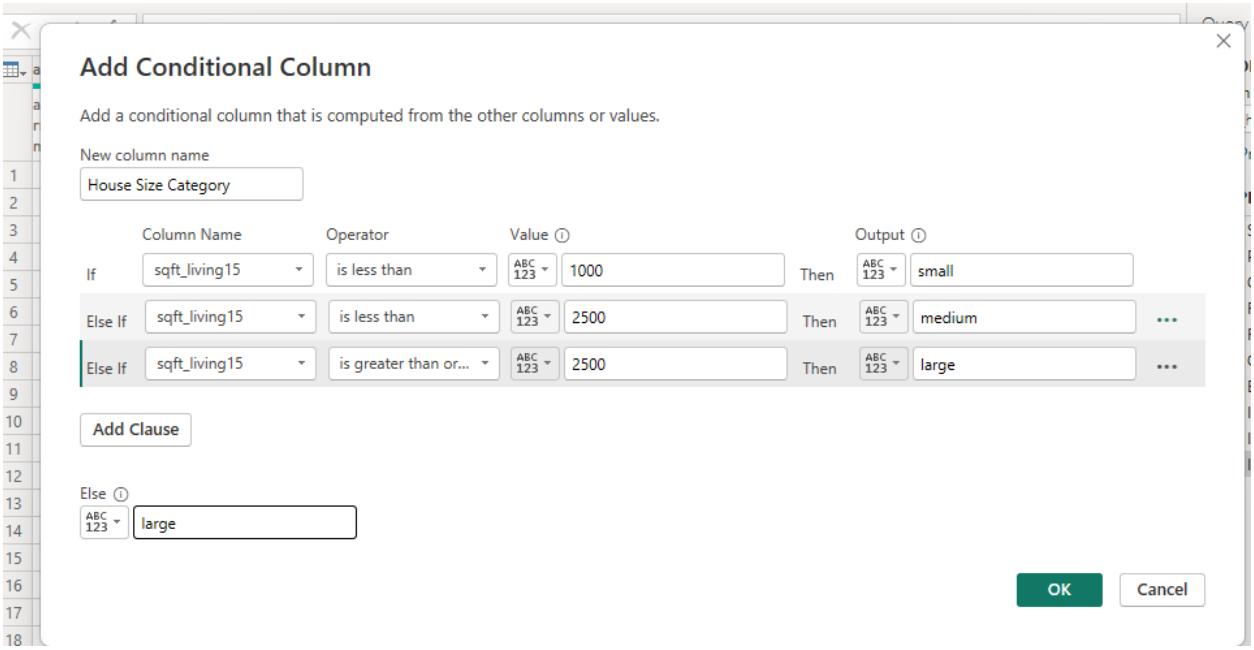
A new conditional column named `House Size Category` was created using the `sqft_living` column.

The following conditions were applied:

Living Area	Category
Below 1000	Small

1000 to below 2500	Medium
2500 and above	Large

This classification helps group properties according to their living area.



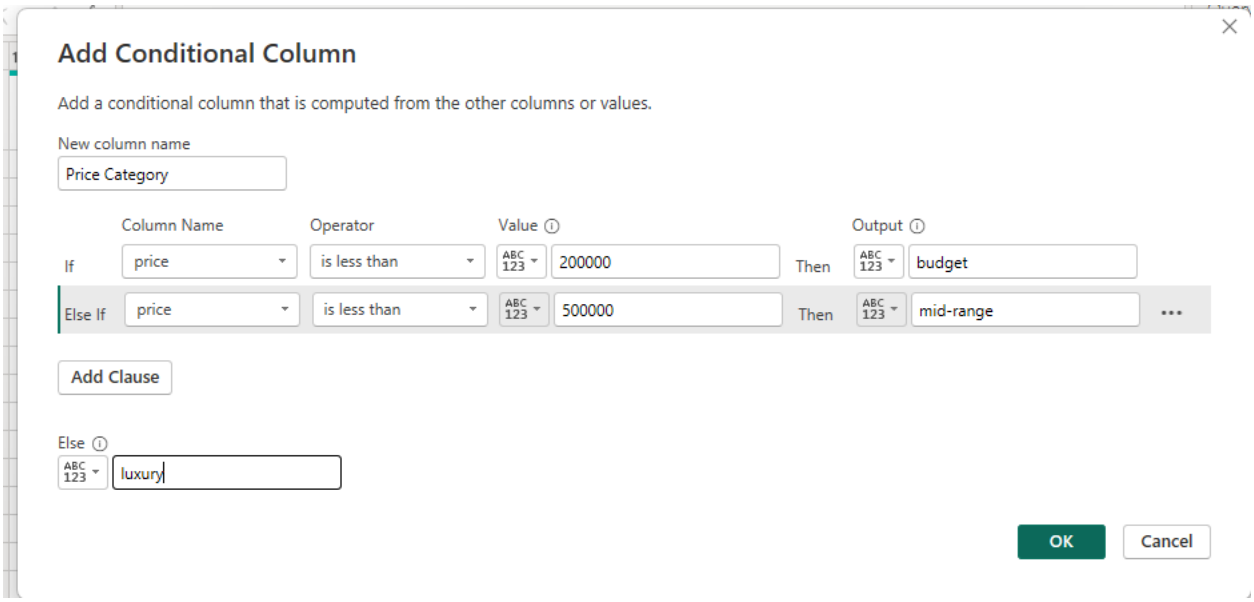
7.2 Price Category

A new conditional column named `Price Category` was created using the `price` column.

The following conditions were applied:

House Price	Category
Below 200,000	Budget

200,000 to below 500,000	Mid-Range
500,000 and above	Luxury



7.3 Waterfront Transformation

The original waterfront column contained numerical values.

The values were replaced with meaningful text labels:

- 0 – No
- 1 – Yes

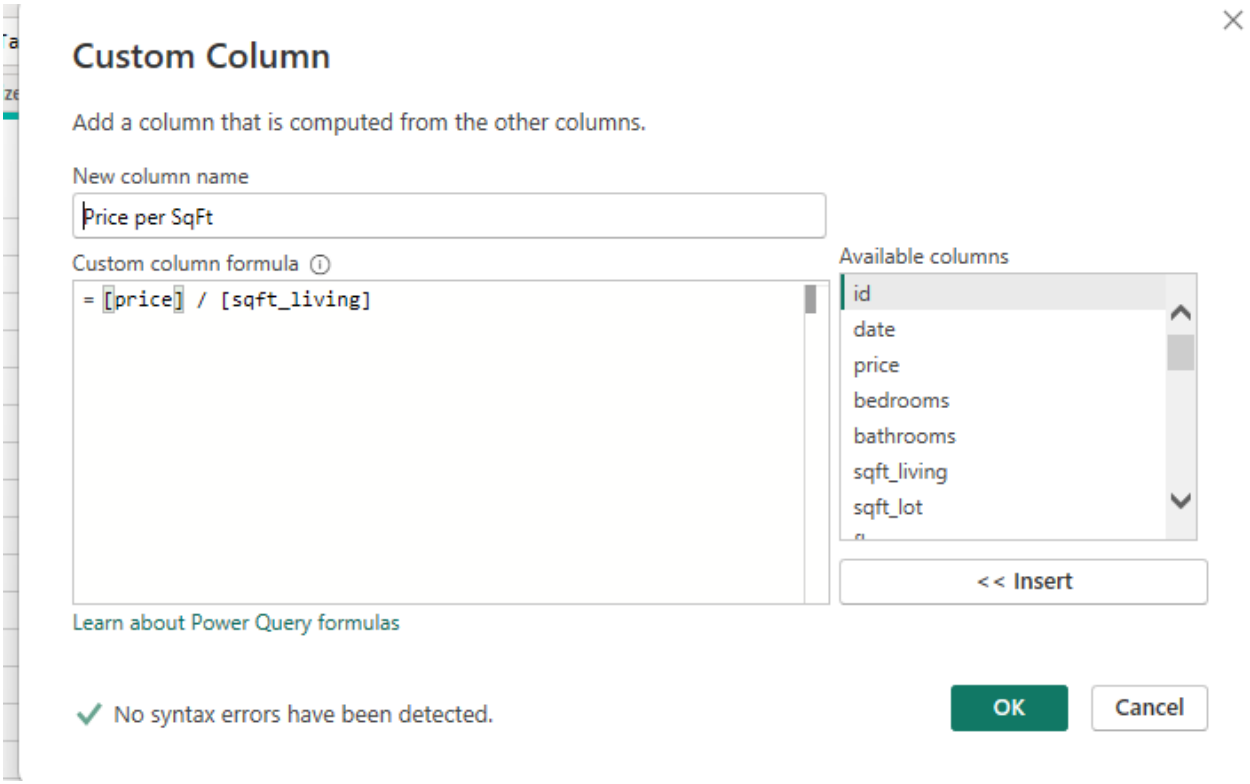
This makes the information easier to understand and interpret.

7.4 View Category

A new conditional column named `View Category` was created using the `view` column.

8. CREATING CUSTOM COLUMNS

A custom column named `Property Age at Sale` was created to calculate the age of the house when it was sold.



11. CONCLUSION

This project provided practical experience in data cleaning and transformation using Microsoft Power BI and Power Query Editor.

Different operations such as removing duplicates, filtering records, extracting date information, creating conditional columns and developing custom calculations were performed.

Through this project, the importance of data cleaning in preparing reliable datasets for analysis was understood.

The cleaned housing dataset can be used for further analysis and for developing an interactive Real-Estate Analytics Dashboard.